Stony Brook Institute at Anhui University

安徽大学纽约石溪学院

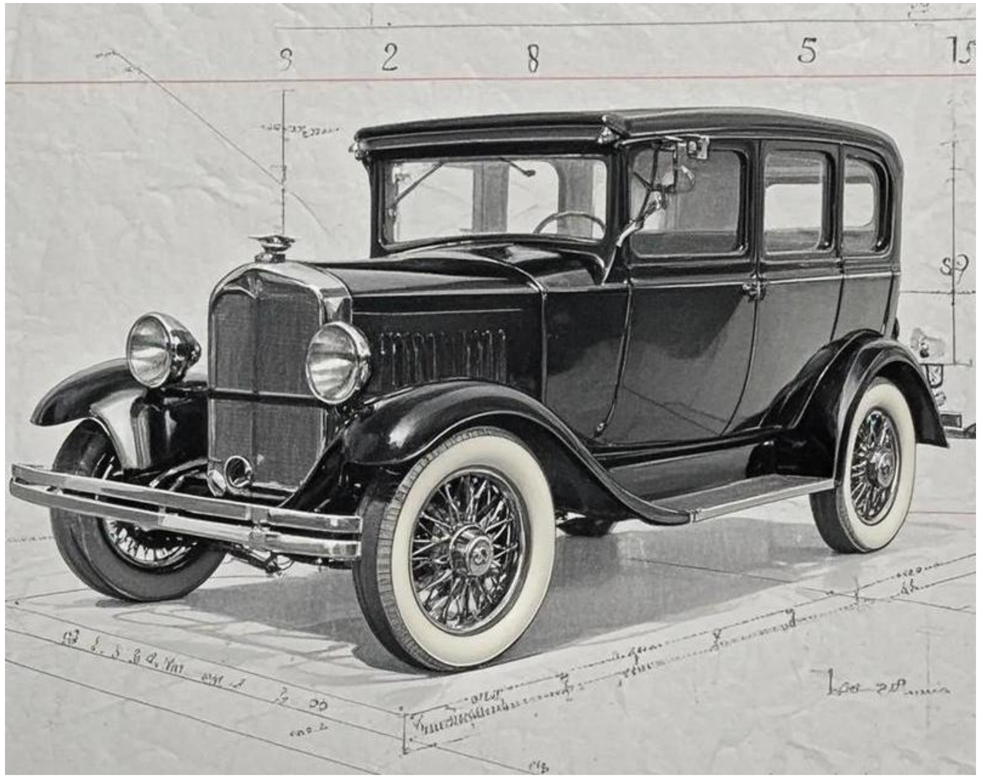
ISE 333: User Interface Development (Design)
Chapter 12: Advancing the User Experience

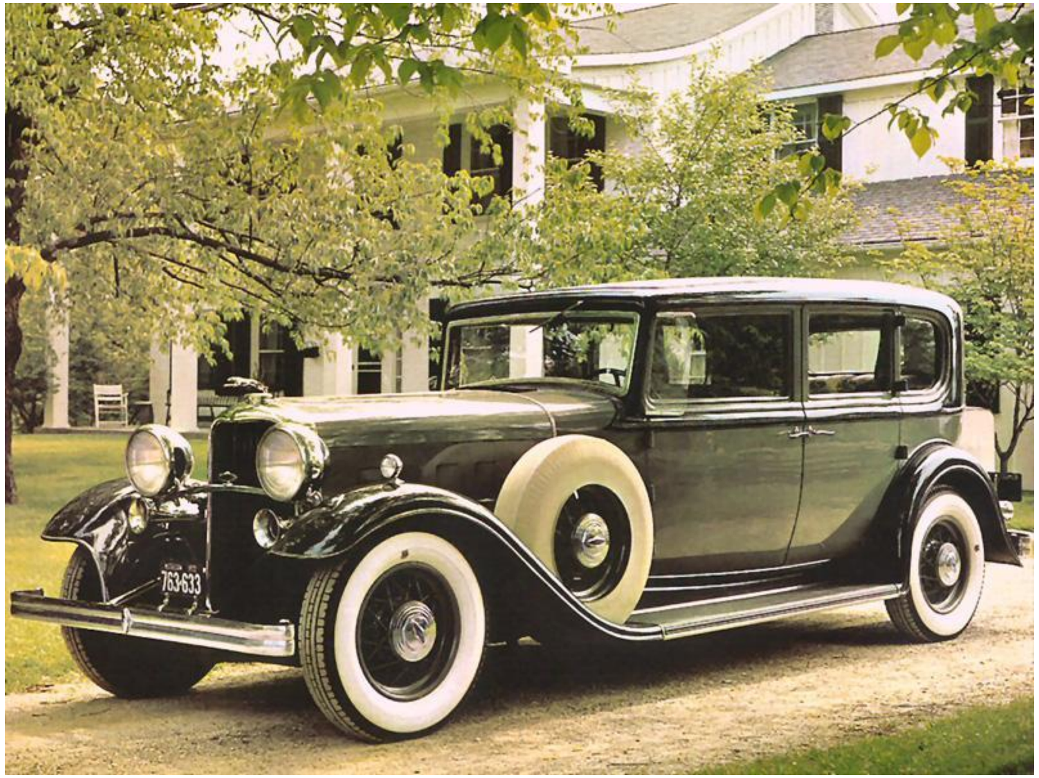
Spring 2026

Instructor: Hao Li

Introduction

- Early Negligence of User Experience (UX)
  - Early automobiles were purely functional
  - Henry Ford could joke about customers getting any colour as long as it was black





Introduction

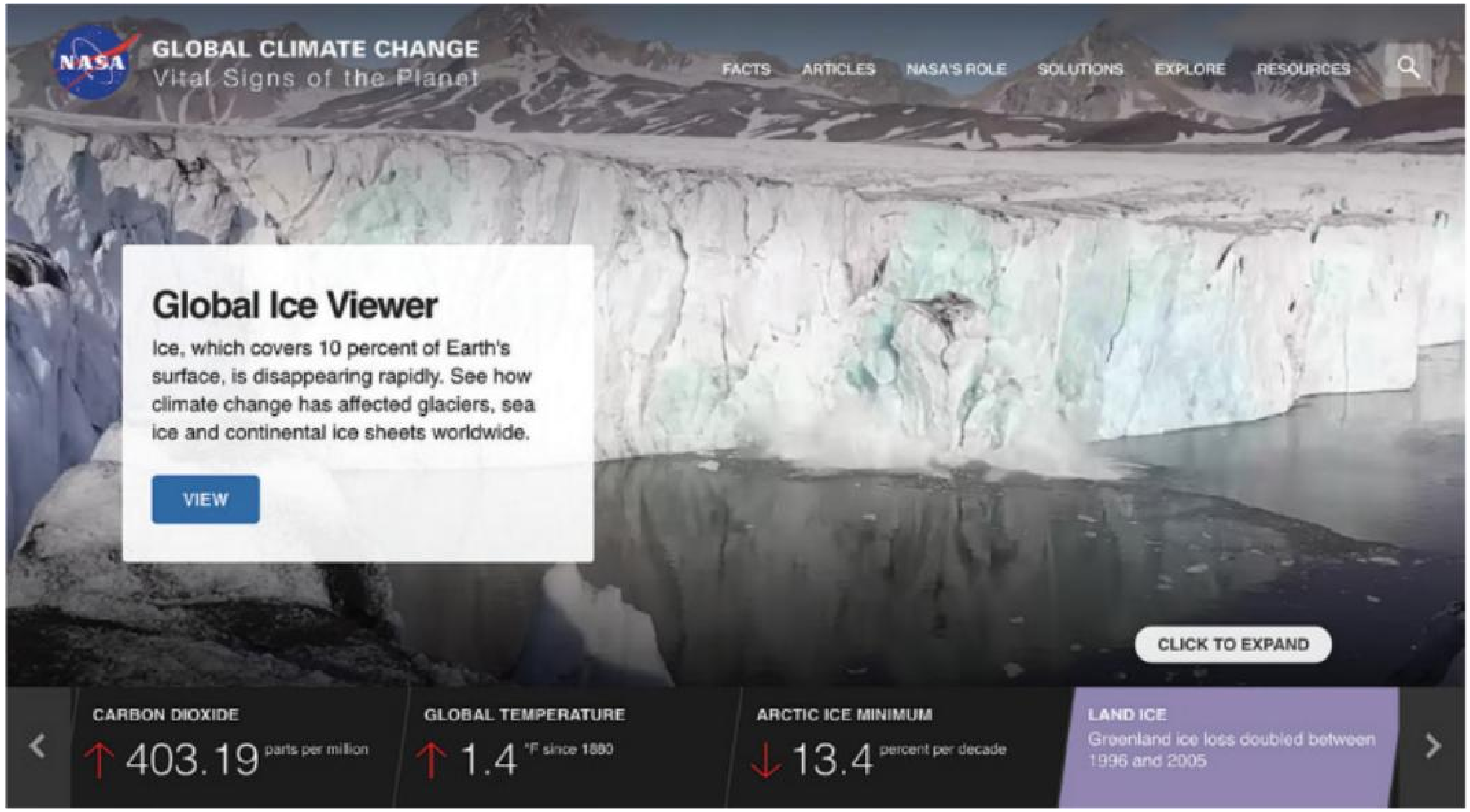
- Nowadays Highly-Valued User Experience (UX)
  - Modern car designers have learned to balance function and fashion

Introduction

- Display Design
- View (Window) Management
- Animation
- Webpage Design
- Colour
- Error Messages

Display Design

- Key Role
  - For most interactive systems, the displays are a key component of successful UIDs
  - The displays are the source of many lively arguments as well



Display Design

- Guidelines and Principles
  - S. Smith, J. Mosier, "Guidelines for designing user interface software", 1986
  - J. Johnson, "Designing with the mind in mind: Simple guide to understanding user interface design rules", 2014

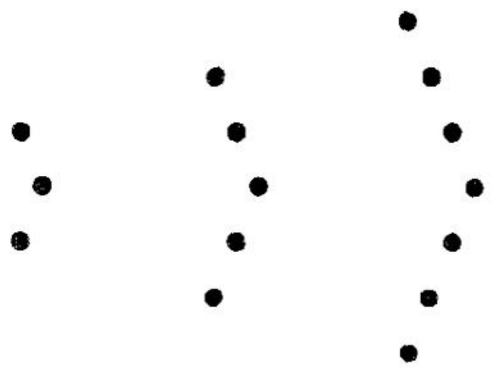
Examples of the 162 data-display guidelines from Smith and Mosier (1986)

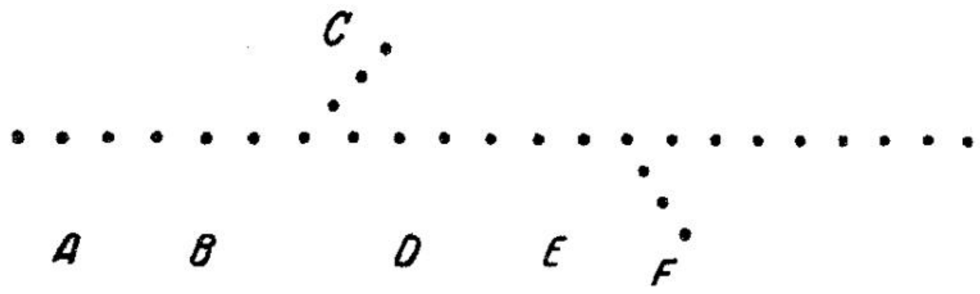
- Ensure that any data that a user needs, at any step in a transaction sequence, are available for display.
- Display data to users in directly usable forms; do not require that users convert displayed data.
- Maintain a consistent format for any particular type of data display from one display to another.
- Use short, simple sentences.
- Use affirmative statements rather than negative statements.
- Adopt a logical principle by which to order lists; where no other principle applies, order lists alphabetically.
- Ensure that labels are sufficiently close to their data fields to indicate association yet are separated from their data fields by at least one space.
- Left-justify columns of alphabetic data to permit rapid scanning.
- Label each page in multi-paged displays to show its relation to the others.
- Begin every display with a title or header, describing briefly the contents or purpose of the display; leave at least one blank line between the title and the body of the display.
- For size coding, make larger symbols be at least 1.5 times the height of the next-smaller symbol.
- Consider color coding for applications in which users must distinguish rapidly among several categories of data, particularly when the data items are dispersed on the display.
- For a large table that exceeds the capacity of one display frame, ensure that users can see column headings and row labels in all displayed sections of the table. Note that one study at the University of Texas published guidelines for tables to reduce any patient safety impacts when using large, tabular displays of patient data (University of Texas, 2013).
- Provide a means for users (or a system administrator) to make necessary changes to display functions, as data-display requirements may change (as is often the case).

Display Design

- Representative Guidelines
  - UIDers should begin with a thorough knowledge of the users’ tasks
  - Effective display designs must provide all the necessary data in the proper sequence to carry out the task
  - Meaningful groupings of items (with labels suitable to the users’ knowledge), consistent sequence of groups, and orderly formats support task performance
  - The *Gestalt Laws of Perception* (rules of organization of perceptual scenes) apply

M. Wertheimer. Untersuchungen zur lehre von der gestalt. ii. *Psychologische Forschung*, 4:301–350, 1923.





Display Design

- Representative Guidelines
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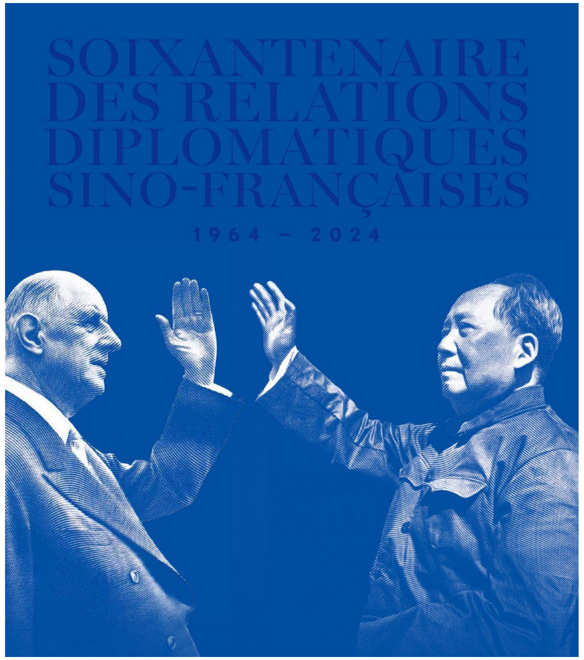
  - Effective display designs must provide all the necessary data in the proper sequence to carry out the task
  - Meaningful groupings of items (with labels suitable to the users' knowledge), consistent sequence of groups, and orderly formats support task performance
  - The *Gestalt Laws of Perception* (rules of organization of perceptual scenes) apply
  - Groups can be surrounded by blank spaces or boxes
  - Related items can (alternatively) be indicated by highlighting, background shading, colour, or special fonts
  - Orderly formats within a group can be accomplished by left or right justification, alignment on decimal points for numbers, or markers to decompose lengthy fields

Display Design

- Representative Principles
  - Graphic designers, originally working in a world of print media, adapted principles for display design



人民日報

中國人民解放戰爭勝利萬歲！

中華人民共和國萬歲！

完成勝利·鞏固勝利

一九五零年元旦獻詞

中國共產黨中央委員會發表告前綫將士和全國同胞書

歡慶一九四九年所獲得的偉大歷史勝利

提出一九五零年光榮的戰鬥與生產任務

解放軍抵昆明以東

人民維新 張瀾

一切為着人民 李濟深

人民日報

RENMIN RIBAO

毛主席就最近日本反美大示威发表谈话

中国人民支持日本人民伟大的爱国斗争

中国和法国决定建立外交关系

两国政府商定在三个月内任命大使

20 PAGES

Le Monde

DANS UN COMMUNIQUÉ CONJOINT

PARIS ET PÉKIN ANNONCENT OFFICIELLEMENT L'ÉTABLISSEMENT DE RELATIONS DIPLOMATIQUES

RENCONTRE de deux indépendances

Formose paraît pencher pour la rupture avec la France

TROIS SIÈCLES DE RELATIONS FRANCO-CHINOISES

I – La France partie prenante dans la conquête de la Chine

Par ROGER PELISSIER

AVERTISSEMENT DE MOSCOU A LONDRES contre une intervention à Zanzibar

• Les troupes britanniques ont rétabli l'ordre en Afrique orientale

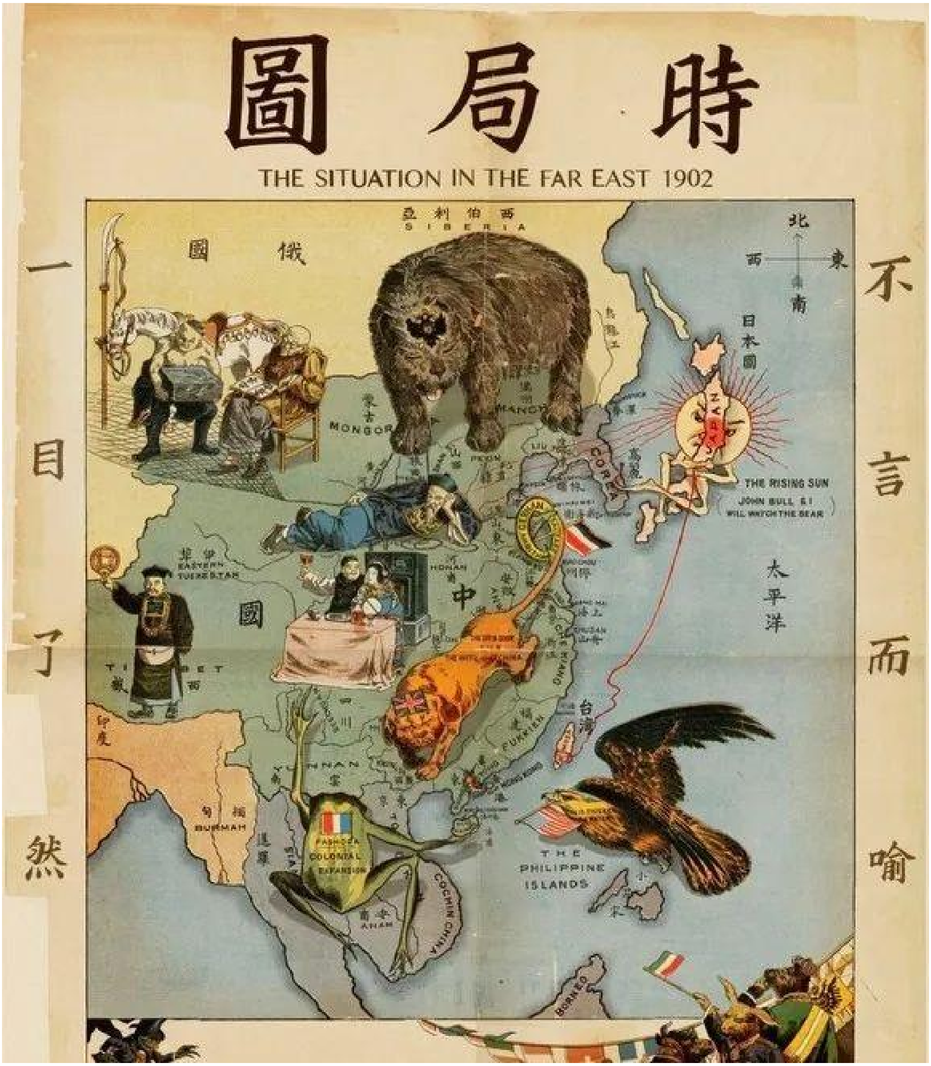
• Nouveaux assassinats de missionnaires au Congo ex-belge

Display Design

- Representative Principles
  - Elegance and simplicity
    - Unity, refinement, and fitness

Display Design

- Representative Principles
  - Elegance and simplicity
    - Unity, refinement, and fitness
  - Scale, contrast, and proportion
    - Clarity, harmony
  - Organization and visual structure
    - Grouping, hierarchy, relationship, and balance
  - Module and program
    - Focus, flexibility, and consistent application
  - Image and representation
    - Immediacy, generality, cohesiveness, and characterization
  - Style
    - Distinctiveness, integrity, comprehensiveness, and appropriateness



Display Design

- Field Layout
  - This example record may contain the necessary information for a task, but extracting the information will be slow and error-prone

Poor:	TAYLOR,SUSAN	34787331	WILLIAM TAYLOR
	THOMAS	10292014	
	ANN	08212015	
	ALEXANDRA	09082012	

  - The children’s names can be listed in chronological order, and familiar separators for the dates aid recognition
  - Consistency (say, for the first/last names order) is important

Better:	SUSAN TAYLOR	34787331	WILLIAM TAYLOR
	ALEXANDRA	09-08-2012	
	THOMAS	10-29-2014	
	ANN	08-21-2015	

Display Design

- Field Layout
  - The children's names can be listed in chronological order, and familiar separators for the dates aid recognition
  - Consistency (say, for the first/last names order) is important

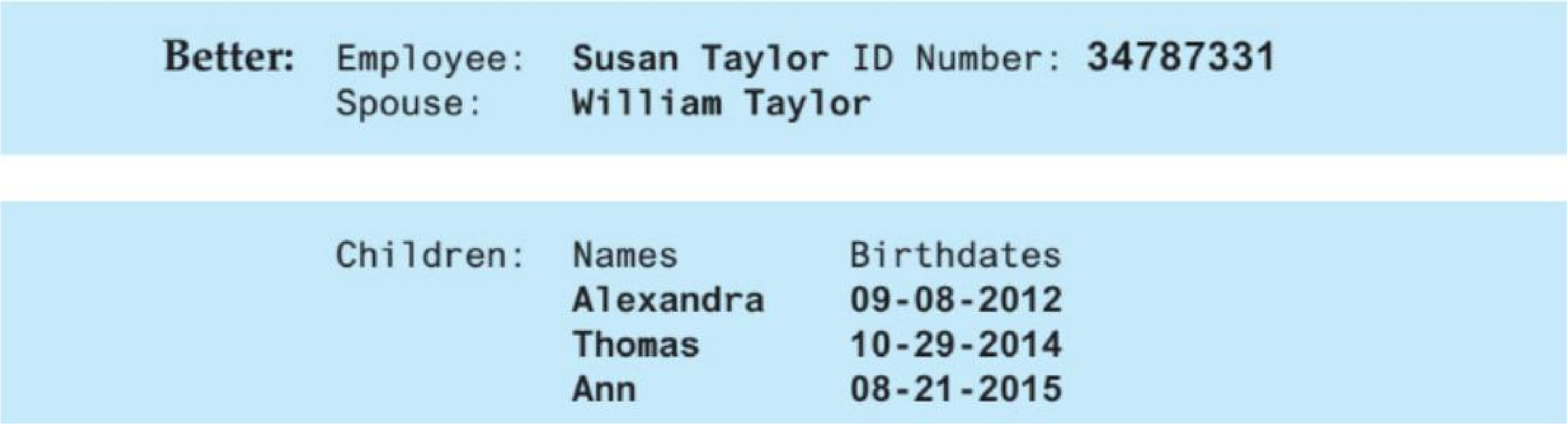
```
Better: SUSAN TAYLOR        34787331        WILLIAM TAYLOR
        ALEXANDRA           09-08-2012
        THOMAS              10-29-2014
        ANN                 08-21-2015
```

  - For most users, labels will be helpful
  - Use boldface and mixed upper- and lowercase for the contents

```
Better:  Employee:  Susan Taylor  ID Number: 34787331
         Spouse:    William Taylor
         Children:  Names         Birthdates
                    Alexandra     09-08-2012
                    Thomas        10-29-2014
                    Ann           08-21-2015
```

Display Design

- Field Layout
  - The children’s names can be listed in chronological order, and familiar separators for the dates aid recognition
  - Consistency (say, for the first/last names order) is important
  - For most users, labels will be helpful
  - Use boldface and mixed upper- and lowercase for the contents
  - Logical groupings can be created by using shading or borders to delineate sets of related information



Display Design

- Empirical Results for Professional Environments
  - Expert users can deal with and may in fact prefer dense packing, multiple displays, and highly coded fields

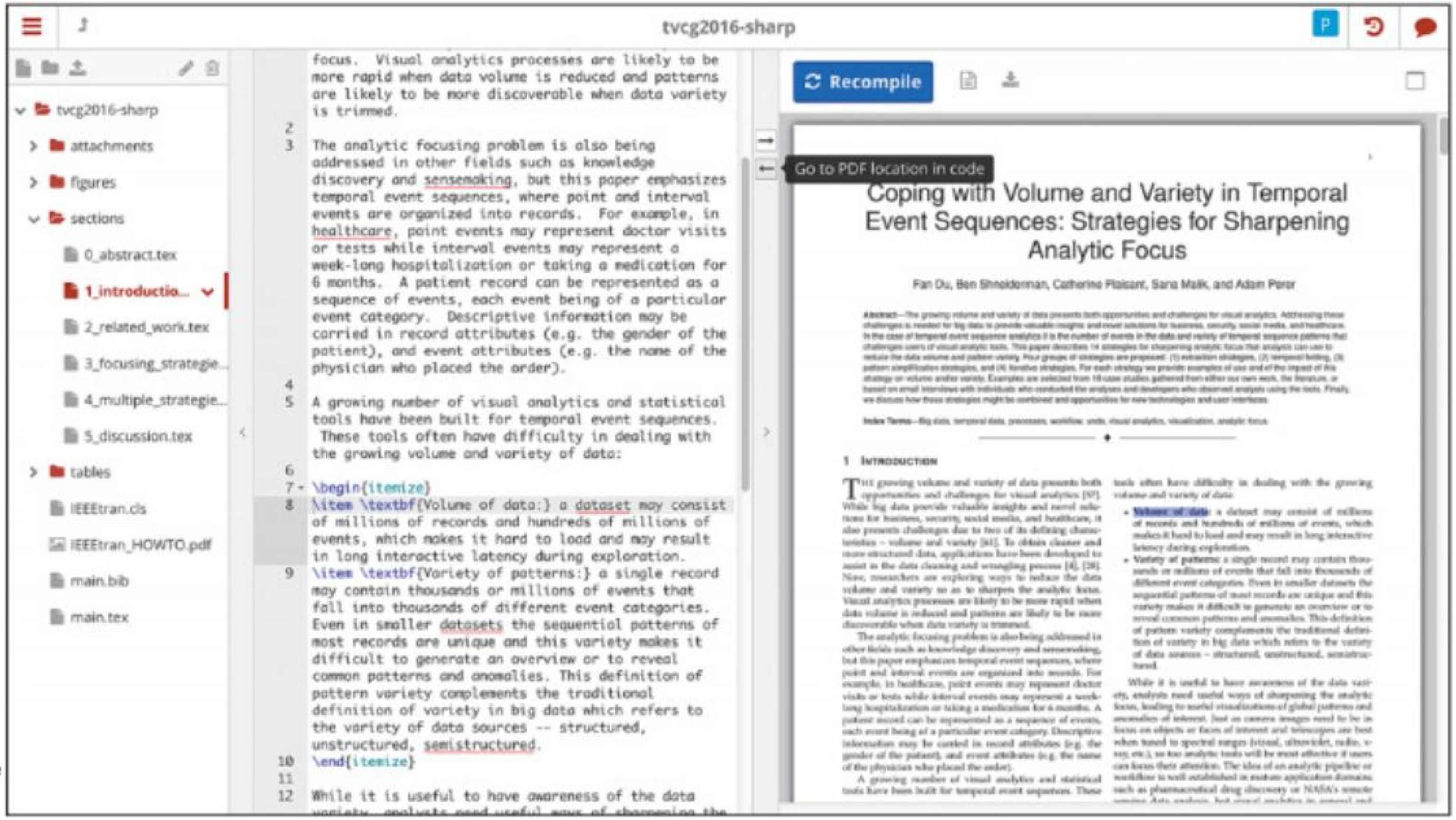
U.S. Navy air-traffic-control work environment, with multiple, specialized, data-intensive displays.

View (Window) Management

- Multiple-Window Display Strategy
  - The visual nature of window use has led many UIDers to apply direct-manipulation strategies to window actions
- Coordinating Multiple Views (Windows)
  - Coordinated windows, i.e., windows that appear, change contents, resize automatically, and close as a direct result of user actions in the task domain
  - Coordination is a task concept that describes how information objects change based on user actions
  - A careful study of user tasks can lead to the development of task-specific coordinations based on sequences of actions

View (Window) Management

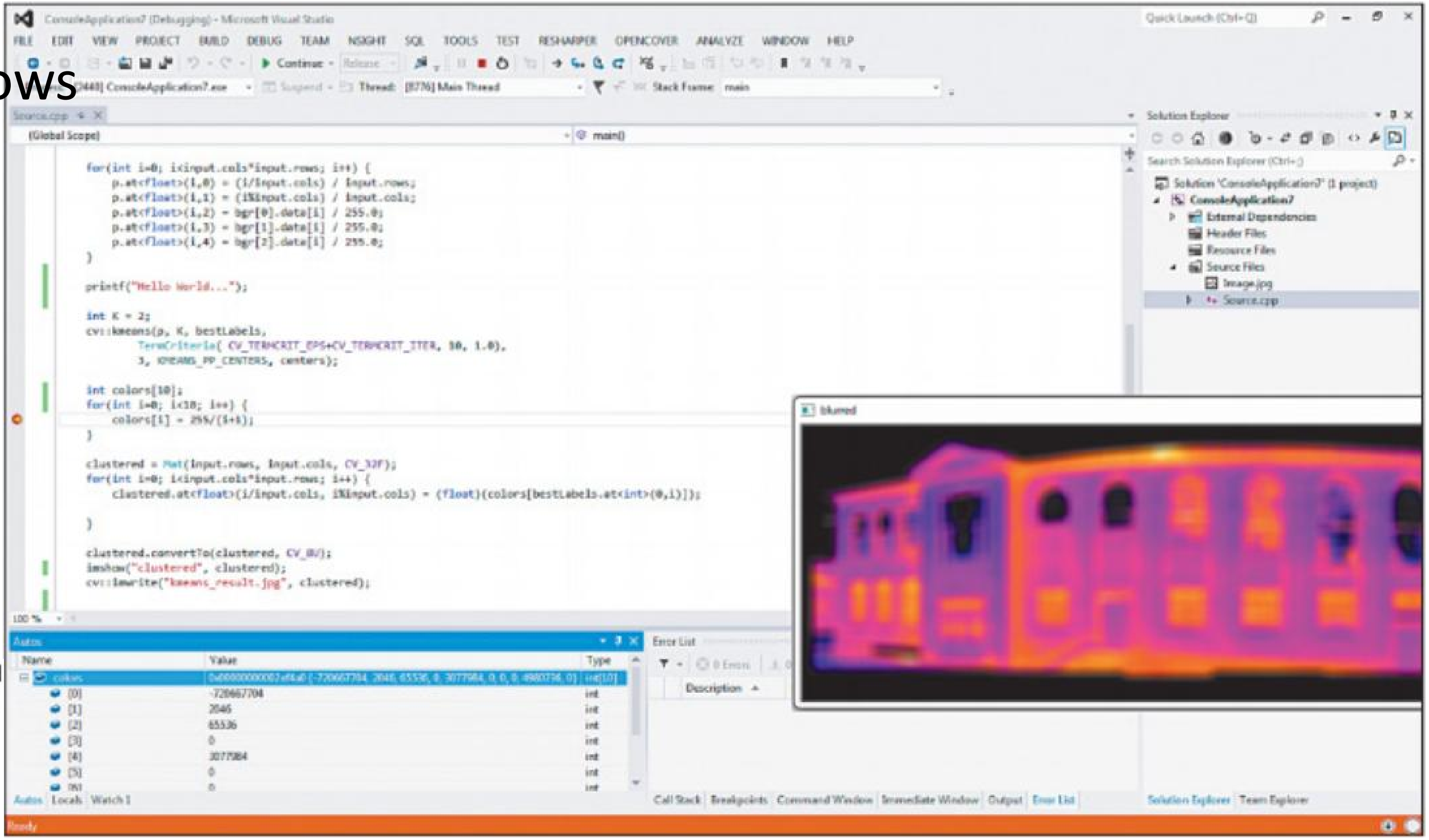
- Coordinating Multiple Views (Windows)
  - Synchronized scrolling
    - Useful for comparing two versions of a program or document
  - Hierarchical browsing



ShareLaTeX allows users to edit a structured LaTeX document and see the resulting formatted document. On the left is the hierarchical list of document sections. The "1. Introduction" section is selected and highlighted in red, and its text can be edited in the middle. The preview of the output is shown on the right. After selecting a passage in one view, it is possible to see the corresponding location on the other view.

View (Window) Management

- Coordinating Multiple Views (Windows)
  - Synchronized scrolling
  - Hierarchical browsing
  - Tiled or overlapping windows



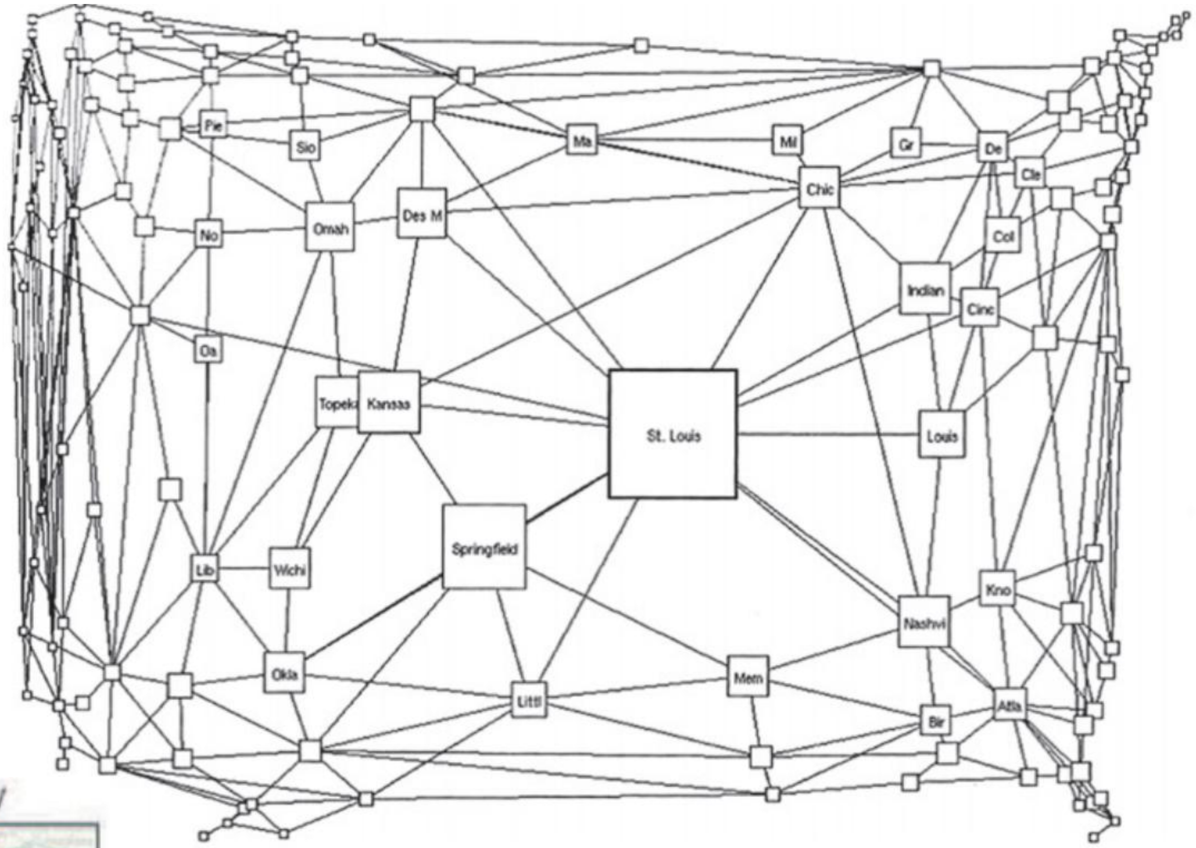
Example of Visual Studio Integrated Development Environment (IDE) illustrating coordinated window views. The project files are listed on the right in a hierarchical browser. The selected file (i.e., "source .cpp") is highlighted in the list and displayed on the left. A breakpoint was set in the code (red dot) at the line starting with "color[i]" so the bottom left window shows the values of the color array of the breakpoint. All windows are titled, except for the output window (the colorful image of heat sensor data), which overlaps all the other windows.

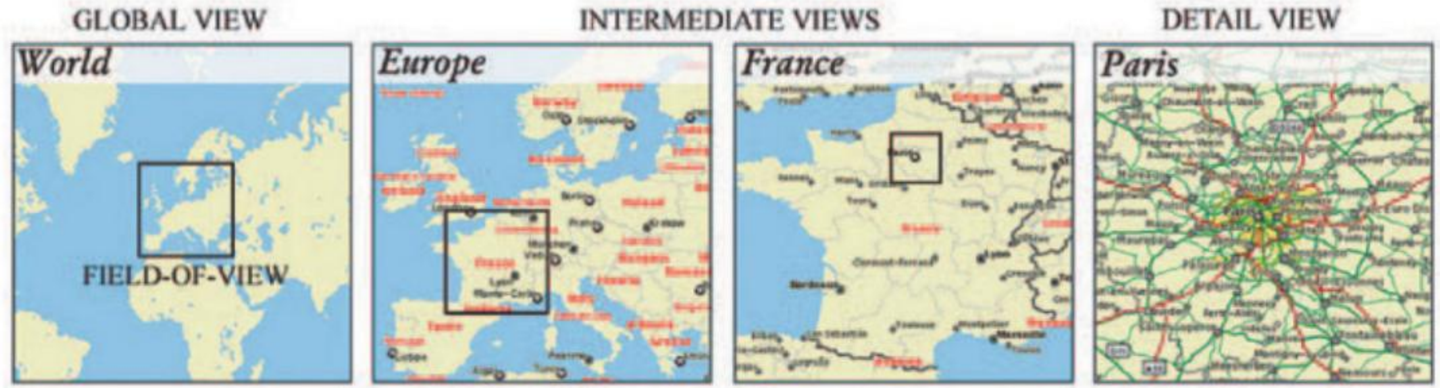
View (Window) Management

- Coordinating Multiple Views (Windows)
  - Synchronized scrolling
  - Hierarchical browsing
  - Tiled or overlapping windows
  - Opening/closing of dependent windows
  - Saving of window state
  - Tabbed browsing

View (Window) Management

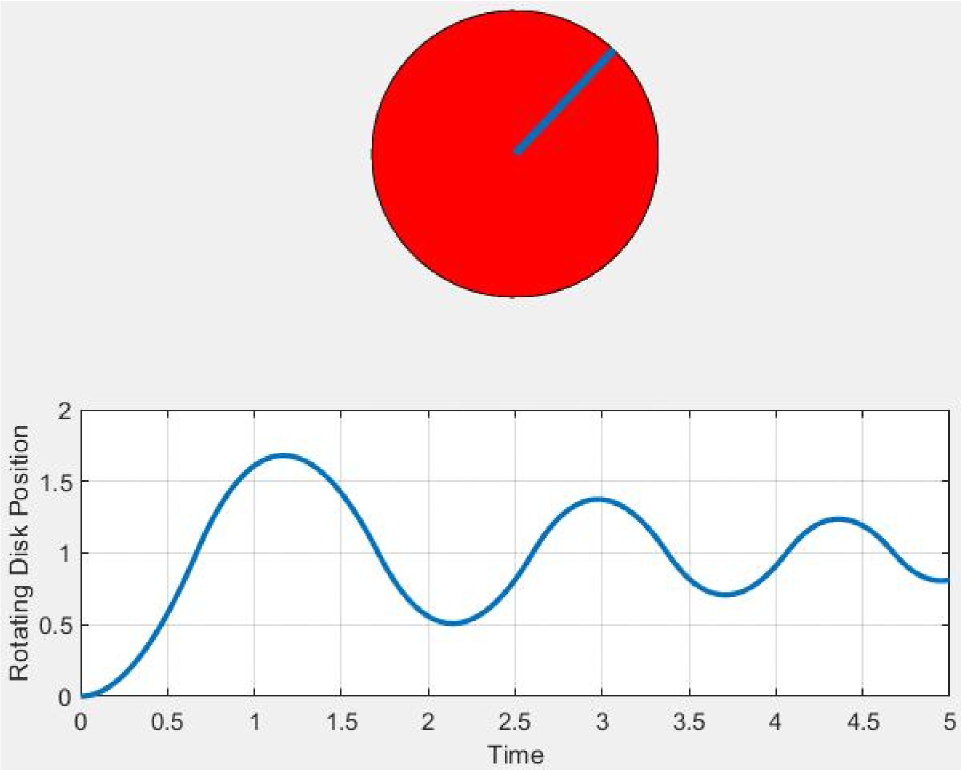
- Browsing Large Views
  - Representative applications
    - Large maps, circuit diagrams, artwork
  - Interaction
    - Overviews (context)
    - Detail views (focus)
- Zoom Views
- Fishseye Views

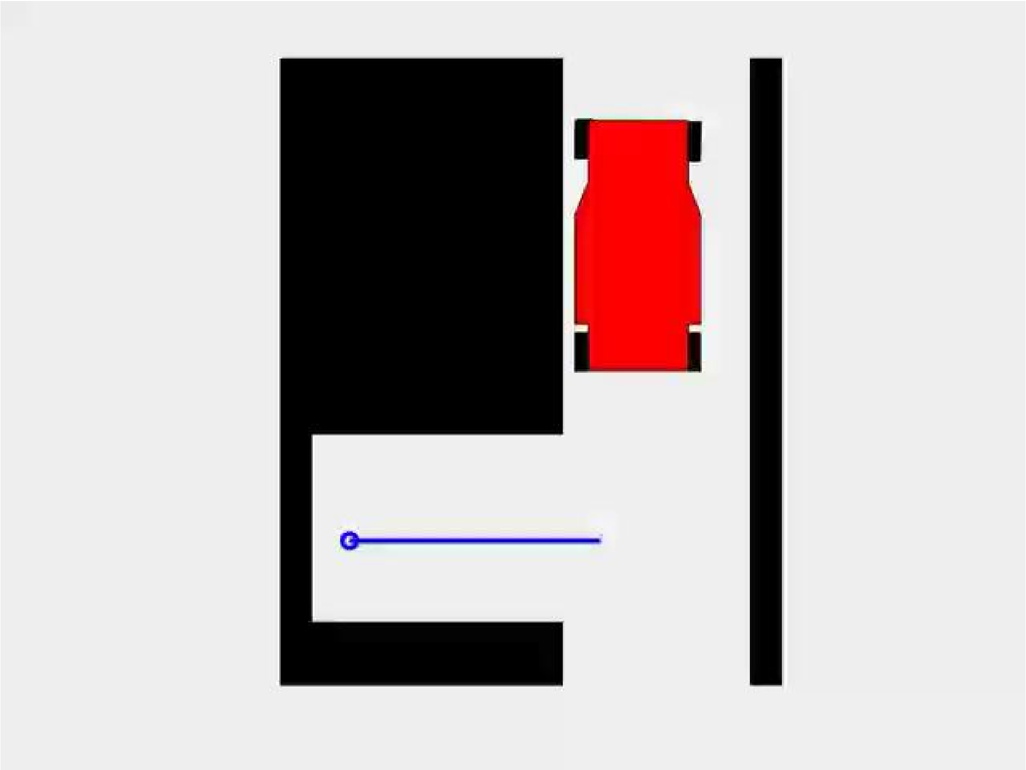




Animation

- Naturalness of Animation
  - Perceiving and interpreting motion is a fundamental element of human perception, and our eyes are attracted to moving objects in the real world
  - Smooth animations are expected to keep users oriented during UI interaction





Webpage Design

- Avoiding Top Mistakes

Top 10 Mistakes

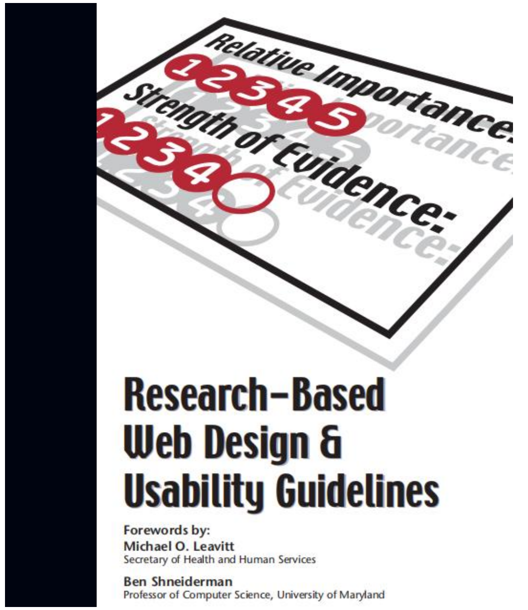
1. Burying information too deep in a website
2. Overloading pages with too much material
3. Providing awkward or confusing navigation
4. Putting information in unexpected places on the page
5. Not making links obvious and clear
6. Presenting information in bad tables
7. Making text so small that many users cannot read it
8. Using color combinations for text that many users cannot read
9. Using bad forms
10. Hiding (or not providing) features that could help users

Tullis, T. S., Information presentation, in Proctor, R., and Vu, K. (Editors), *Handbook of Human Factors in Web Design*, Routledge, New York (2004).

Tullis, T. S., Web-based presentation of information: The top ten mistakes and why they are mistakes, *Proceedings of the HCI International 2005*, Las Vegas, NV, Lawrence Erlbaum Associates (July 2005).

Webpage Design

- Guidelines



Guideline Heading	Relative Importance
Provide Useful Content	5
Establish User Requirements	5
Understand and Meet User's Expectations	5
Involve Users in Establishing User Requirements	5
Do Not Display Unsolicited Windows or Graphics	5
Comply with Section 508	5
Design Forms for Users Using Assistive Technology	5
Do Not Use Color Alone to Convey Information	5
Enable Access to the Homepage	5
Show All Major Options on the Homepage	5
Create a Positive First Impression of Your Site	5
Avoid Cluttered Displays	5
Place Important Items Consistently	5
Place Important Items at Top Center	5
Eliminate Horizontal Scrolling	5
Use Clear Category Labels	5
Use Meaningful Link Labels	5
Distinguish Required and Optional Data Entry Fields	5
Label Pushbuttons Clearly	5
Make Action Sequences Clear	5
Organize Information Clearly	5
Facilitate Scanning	5
Ensure that Necessary Information is Displayed	5
Ensure Usable Search Results	5
Design Search Engines to Search the Entire Site	5
Set and State Goals	4
Focus on Performance Before Preference	4
Consider Many User Interface Issues	4
Be Easily Found in the Top 30	4
Increase Web Site Credibility	4
Standardize Task Sequences	4
Reduce the User's Workload	4
Design For Working Memory Limitations	4
Minimize Page Download Time	4
Warn of 'Time Outs'	4
Display Information in a Directly Usable Format	4
Format Information for Reading and Printing	4
Provide Feedback when Users Must Wait	4
Inform Users of Long Download Times	4
Develop Pages that Will Print Properly	4
Enable Users to Skip Repetitive Navigation Links	4
Provide Text Equivalents for Non-Text Elements	4

Guideline Heading	Relative Importance
Test Plug-Ins and Applets for Accessibility	4
Design for Common Browsers	4
Account for Browser Differences	4
Design for Popular Operating Systems	4
Design for User's Typical Connection Speed	4
Communicate the Web Site's Value and Purpose	4
Limit Prose Text on the Homepage	4
Ensure the Homepage Looks like a Homepage	4
Structure for Easy Comparison	4
Establish Level of Importance	4
Optimize Display Density	4
Align Items on a Page	4
Provide Navigational Options	4
Differentiate and Group Navigation Elements	4
Use a Clickable 'List of Contents' on Long Pages	4
Provide Feedback on Users' Location	4
Place Primary Navigation Menus in the Left Panel	4
Provide Descriptive Page Titles	4
Use Descriptive Headings Liberally	4
Use Unique and Descriptive Headings	4
Highlight Critical Data	4
Use Descriptive Row and Column Headings	4
Link to Related Content	4
Match Link Names with Their Destination Pages	4
Avoid Misleading Cues to Click	4
Repeat Important Links	4
Use Text for Links	4
Designate Used Links	4
Use Black Text on Plain, High-Contrast Backgrounds	4
Format Common Items Consistently	4
Use Mixed-Case for Prose Text	4
Ensure Visual Consistency	4
Order Elements to Maximize User Performance	4
Place Important Items at Top of the List	4
Format Lists to Ease Scanning	4
Display Related Items in Lists	4
Label Data Entry Fields Consistently	4
Do Not Make User-Entered Codes Case Sensitive	4
Label Data Entry Fields Clearly	4
Minimize User Data Entry	4
Use Simple Background Images	4

Guideline Heading	Relative Importance
Ensure that Images Do Not Slow Downloads	4
Use Video, Animation, and Audio Meaningfully	4
Include Logos	4
Graphics Should Not Look like Banner Ads	4
Limit Large Images Above the Fold	4
Ensure Web Site Images Convey Intended Messages	4
Avoid Jargon	4
Use Familiar Words	4
Define Acronyms and Abbreviations	4
Use Abbreviations Sparingly	4
Use Mixed Case with Prose	4
Limit the Number of Words and Sentences	4
Group Related Elements	4
Minimize the Number of Clicks or Pages	4
Make Upper- and Lowercase Search Terms Equivalent	4
Provide a Search Option on Each Page	4
Design Search Around Users' Terms	4
Use an Iterative Design Approach	4
Set Usability Goals	3
Do Not Require Users to Multitask While Reading	3
Use Users' Terminology in Help Documentation	3
Provide Printing Options	3
Ensure that Scripts Allow Accessibility	3
Provide Equivalent Pages	3
Provide Client-Side Image Maps	3
Synchronize Multimedia Elements	3
Do Not Require Style Sheets	3
Design for Commonly Used Screen Resolutions	3
Limit Homepage Length	3
Use Fluid Layouts	3
Avoid Scroll Stoppers	3
Set Appropriate Page Lengths	3
Use Moderate White Space	3
Use Descriptive Tab Labels	3
Present Tabs Effectively	3
Use Headings in the Appropriate HTML Order	3
Provide Consistent Clickability Cues	3
Ensure that Embedded Links are Descriptive	3
Use 'Pointing-and-Clicking'	3
Use Appropriate Text Link Lengths	3
Indicate Internal vs. External Links	3

Guideline Heading	Relative Importance
Link to Supportive Information	3
Use Bold Text Sparingly	3
Use Attention-Attracting Features when Appropriate	3
Use Familiar Fonts	3
Use at Least a 12-Point Font	3
Introduce Each List	3
Use Static Menus	3
Put Labels Close to Data Entry Fields	3
Allow Users to See Their Entered Data	3
Use Radio Buttons for Mutually Exclusive Selections	3
Use Familiar Widgets	3
Anticipate Typical User Errors	3
Partition Long Data Items	3
Use a Single Data Entry Method	3
Prioritize Pushbuttons	3
Use Check Boxes to Enable Multiple Selections	3
Label Units of Measurement	3
Do Not Limit Viewable List Box Options	3
Display Default Values	3
Limit the Use of Images	3
Include Actual Data with Data Graphics	3
Display Monitoring Information Graphically	3
Limit Prose Text on Navigation pages	3
Use Active Voice	3
Write Instructions in the Affirmative	3
Make First Sentences Descriptive	3
Design Quantitative Content for Quick Understanding	3
Display Only Necessary Information	3
Format Information for Multiple Audiences	3
Allow Simple Searches	3
Notify Users when Multiple Search Options Exist	3
Include Hints to Improve Search Performance	3
Solicit Test Participants' Comments	3
Evaluate Web Sites Before and After Making Changes	3
Prioritize Tasks	3
Distinguish Between Frequency and Severity	3
Select the Right Number of Participants	3
Use Parallel Design	2
Provide Assistance to Users	2
Provide Frame Titles	2
Avoid Screen Flicker	2

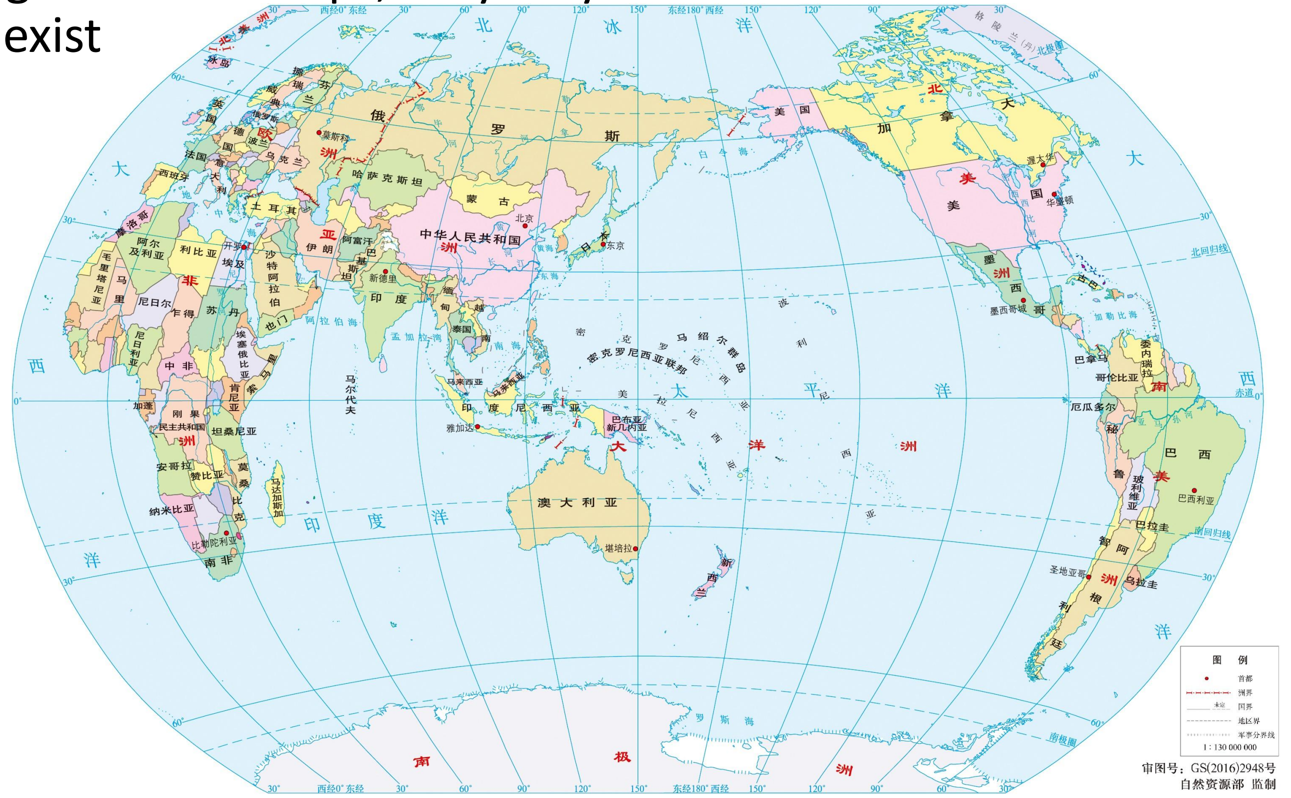
Guideline Heading	Relative Importance
Attend to Homepage Panel Width	2
Choose Appropriate Line Lengths	2
Keep Navigation-Only Pages Short	2
Use Appropriate Menu Types	2
Use Site Maps	2
Facilitate Rapid Scrolling While Reading	2
Use Scrolling Pages for Reading Comprehension	2
Use Paging Rather Than Scrolling	2
Scroll Fewer Screenfuls	2
Provide Users with Good Ways to Reduce Options	2
Color-Coding and Instructions	2
Emphasize Importance	2
Highlighting Information	2
Start Numbered Items at One	2
Use Appropriate List Style	2
Place Cursor in First Data Entry Field	2
Ensure that Double-Clicking Will Not Cause Problems	2
Use Open Lists to Select One from Many	2
Use Data Entry Fields to Speed Performance	2
Use a Minimum of Two Radio Buttons	2
Provide Auto-Tabbing Functionality	2
Introduce Animation	2
Emulate Real-World Objects	2
Use Thumbnail Images to Preview Larger Images	2
Use Color for Grouping	2
Provide Search Templates	2
Use the Appropriate Prototyping Technology	2
Use Inspection Evaluation Results Cautiously	2
Recognize the 'Evaluator Effect'	2
Use Personas	1
Use Frames When Functions Must Remain Accessible	1
Use 'Glosses' to Assist Navigation	1
Breadcrumb Navigation	1
Capitalize First Letter of First Word in Lists	1
Minimize Use of the Shift Key	1
Use Images to Facilitate Learning	1
Using Photographs of People	1
Apply Automatic Evaluation Methods	1
Use Cognitive Walkthroughs Cautiously	1
Choosing Laboratory vs. Remote Testing	1
Use Severity Ratings Cautiously	1

Colour

- Use
  - Soothe or strike the eye
  - Add accents to an uninteresting display
  - Facilitate subtle discrimination in complex displays
  - Emphasize the logical organization of information
  - Draw attention to warnings
  - Evoke strong emotional reactions of joy, excitement, fear, or anger

Colour

- Guidelines Rather Than Rules
  - No simple set of rules governs use of colour
  - Use colour conservatively
    - When the colours do not show meaningful relationships, they may mislead users into searching for relationships that do not exist
    - Misleading even for reasonable usage

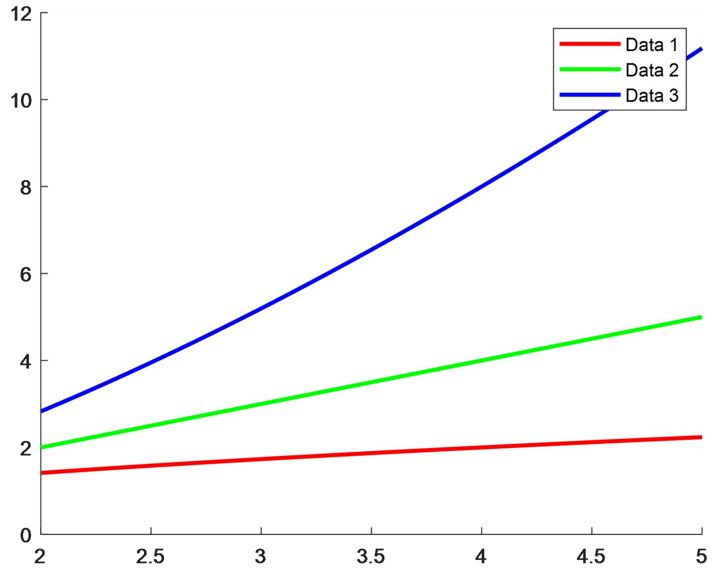


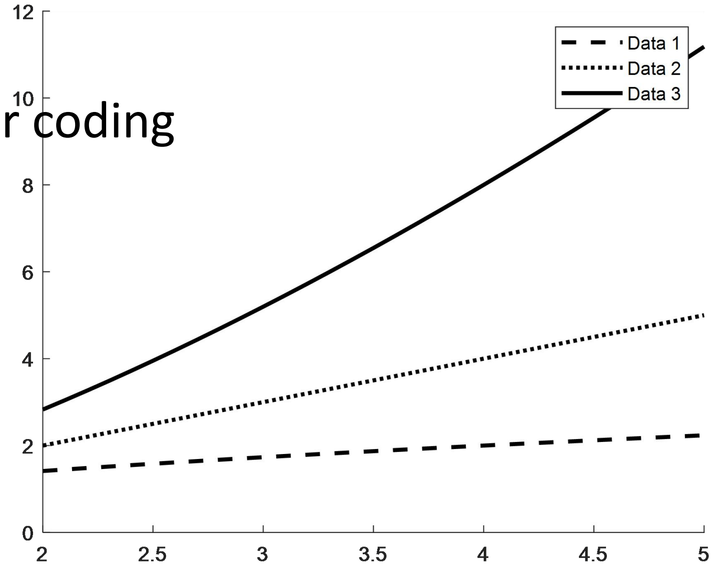
Colour

- Guidelines Rather Than Rules
  - No simple set of rules governs use of colour
  - Use colour conservatively
  - Limit the number of colours
    - A limit of four colours in a single display
    - A limit of seven colours in the entire sequence of displays
  - Recognize the power of colour as a coding technique
    - Colour speeds recognition for many tasks
    - In an air-traffic-control UI, high- and low-flying planes may be colour-coded differently
    - In a programming language IDE, keywords and variables may be colour-coded differently
  - Ensure that colour coding supports the task
  - Have colour coding appear with minimal user effor
    - Users should not have to activate colour coding each time they perform a task; rather, the colour coding should appear automatically

Colour

- Guidelines Rather Than Rules
  - Use colour conservatively
  - Limit the number of colours
  - Recognize the power of colour as a coding technique
  - Ensure that colour coding supports the task
  - Have colour coding appear with minimal user effor
  - Place colour coding under user control
    - When appropriate, the users should be able to turn off colour coding
  - Design/develop for monochrome first
    - Try to resort first to logical patterns instead of colours





Colour

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  - Ensure that colour coding supports the task
  - Have colour coding appear with minimal user effor
  - Place colour coding under user control
  - Design/develop for monochrome first
  - Consider the needs of colour-deficient users
  - Be consistent in colour coding
  - Be alert to common or cultural expectations about colour codes
  - Use colour changes to indicate status changes

Error Messages

- Key Role
  - Error messages are a key pa
- Representative Guidelines

End product

- Be as specific and precise as possible. Determine necessary, relevant error messages.
- Be constructive. Indicate what the user needs to do.
- Use a positive tone. Avoid condemnation. Be courteous.
- Choose user-centered phrasing. State the problem, cause, and solution.
- Consider multiple levels of messages. State brief, sufficient information to assist with the corrective action.
- Maintain consistent grammatical forms, terminology, and abbreviations.
- Maintain consistent visual format and placement.

Development process

- Increase attention to message design.
- Establish quality control.
- Develop and enforce guidelines.
- Carry out usability tests.
- Consider conducting "error handling" reviews.
- Record the frequency of occurrence for each message.

Error Messages

- Representative Guidelines
  - Specificity
    - Messages that are too general make it difficult for the novice to determine what has gone wrong
    - The right amount of specificity is important

Poor: `SYNTAX ERROR`
Better: `Unmatched left parenthesis`

Poor: `INVALID DATA`
Better: `Days range from 1 to 31`

Poor: `BAD FILE NAME`
Better: `The file C:\demo\data.txt was not found`

Poor: `???`
Better: `Touch icon twice to start app`

Error Messages

- Representative Guidelines
  - Specificity
    - The right amount of specificity is important
  - Constructive Guidance
    - Refrain from purely condemning users for what have gone wrong
    - Indicate what users need to do to set things right

Poor: SYNTAX ERROR
Better: Unmatched left parenthesis

Poor: INVALID DATA
Better: Days range from 1 to 31

Poor: BAD FILE NAME
Better: The file C:\demo\data.txt was not found

Poor: ???
Better: Touch icon twice to start app

Poor: Run-Time error '-2147469 (800405)': Method 'Private Profile String' of object 'System' failed.
Better: Virtual memory space consumed. Close some programs and retry.

Poor: Network connection refused.
Better: Password was not recognized. Please retype.

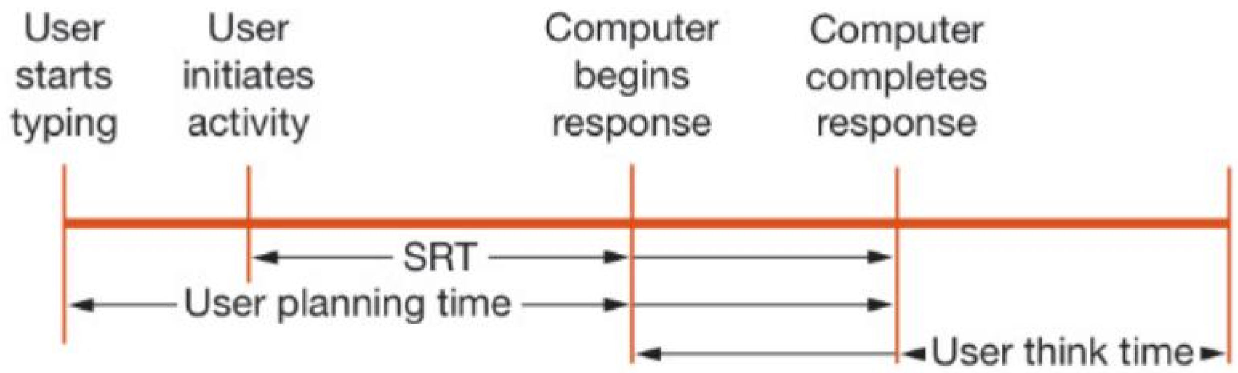
Poor: Invalid date.
Better: Arrival date must come after departure date.

Timely User Experience

- System Response Time
  - For users, the main user experience is the *system response time* (*SRT*, also referred to simply as *response time*)
- Related Human Factors
  - Time delay
    - People cannot perceive any difference in event times less than 25 milliseconds
    - People have trouble perceiving short event times until they approach 100 milliseconds
  - Reaction time
    - It varies for each user, yet usually at a sub-second level
    - In practice users do not seem to be bothered much by one-second delays in changing screens for PC applications

Timely User Experience

- Models of System Response Time



- Empirical Knowledge
  - Users prefer rapid interactions, though they may accept slower responses (latency) for some tasks (e.g., calculations)
  - Lengthy (longer than 5 seconds) response times are generally detrimental to productivity, increasing error rates and decreasing satisfaction
  - More rapid (less than 1 second) interactions are generally preferred and increase productivity, but may also increase error rates for complex tasks

Timely User Experience

- Expectations and Attitudes
  - How long will users wait for the UI to respond before they become annoyed?
  - Previous experiences
    - The first factor influencing acceptable SRT is users' past experiences based on which they have established expectations of the time required to complete a task
  - Individual personality differences
    - The second factor is the individual's tolerance for delays, which varies largely due to many factors such as personality, cost, age, mood, cultural context, time of day, noise, and perceived pressure to complete work
  - Task differences
    - The third factor is the task complexity as well as users' familiarity with the task

DISCUSSION

Q & A